CLASS 9 MATHEMATICS • CHAPTER 7

The Mathematics of Maybe: Introduction to Probability

Exhaustive point-by-point presentation covering every concept, scale, historical origin, and probability model directly from the textbook.

✦ Akash Srivastva (www.akashmaths.online)

Probability: The Measurement of Likelihood

MEASURING THE INTANGIBLE

Just as geometry measures physical space (length, area, volume), probability measures the likelihood or certainty of an event occurring in the presence of uncertainty.

Random Events

Situations where all possible outcomes are known, but the exact outcome cannot be known in advance:

- Will it rain today? (Rain or No rain)
- Will our hockey team win, draw, or lose?
- Will your name be picked in the assembly draw?

Subjective Probability

Decisions based on personal perception or partial clues:

- Friend A: *"The sun is bright, so rain is unlikely."*
- Friend B: *"It's very hot and humid, so rain might come later."*
- Subjective estimates vary between observers based on their personal assessment.

The Need for Objectivity

Moving from intuition to mathematics:

- Modern decision-making requires objective evidence rather than subjective guesswork.
- We need systematic tools to quantify risk in science, business, meteorology, and society.

Classroom Check: Likelihood & Uncertainty

1. How is measuring probability fundamentally similar to measuring physical quantities like length or area?

   Think about what kind of quantity each measures.

2. Why is a monthly assembly lucky draw considered a random event?

   Consider whether all outcomes are known and whether any slip has a special advantage.

3. Give an everyday example of a statement that expresses subjective probability rather than objective probability.

   Recall the conversation between friends about the weather.

4. Can an outcome in a truly random experiment ever be predicted with 100% certainty before the trial occurs?

   Reflect on the definition of chance.

The Anatomy of a Random Observation

DEFINITION: RANDOM EXPERIMENT (TRIAL)

A repeatable process or action where all possible outcomes are known beforehand, but the specific outcome on any single trial is strictly unpredictable.

Classic Experiments

- **Coin Toss:** Possible outcomes are {Heads, Tails}. In a single toss, you cannot be certain which will appear.
- **Die Roll:** Possible outcomes are {1, 2, 3, 4, 5, 6}. Each individual face is unpredictable.
- We can state what c*ould* happen, never what *will* happen!

Fairness in Sports

Think and Reflect (Page 156):

- Why is a coin toss universally used to decide who bats first in a cricket match?
- Because it is completely unbiased: neither captain can manipulate the result, and both sides have an exact 50% chance.

Why is Weather Random?

Complex Sensitivity (Page 157):

- Atmospheric systems depend on countless sensitive variables: temperature, air pressure, humidity, and wind currents.
- Tiny variations multiply exponentially, making rainfall unpredictable with total certainty.

Classroom Check: Experiments & Unpredictability

1. If you toss a ₹1 coin, your friend can guess either heads or tails. Why can they never know the outcome for certain?

   Contrast knowing all possibilities with knowing the actual result.

2. Why would rolling a weighted or uneven die fail to qualify as a fair random experiment?

   Think about equal chances for all faces.

3. Why do meteorologists speak of a "40% chance of rain" rather than claiming it will definitely rain in a specific neighborhood?

   Connect this to atmospheric complexity.

4. Can an experiment be called a "random experiment" if it cannot be repeated under identical conditions?

   Review the textbook definition of a trial.

The Probability Continuum: From 0 to 1

0.0 (Impossible) — ↓0.25 (Less Likely) — ↔0.50 (Even Chance) — ↑0.75 (More Likely) — 1.0 (Certain)



Boundary: P = 0

Impossible Event: Can never occur under any circumstance.

- Rolling a 7 on a standard 6-sided die.
- Winning a sports match without ever playing it.
- It snowing in Mumbai in July!

Benchmark: P = 0.5

Even Chance (Equally Likely): Occurrence and non-occurrence have identical likelihood.

- Getting Heads on a fair coin toss (50%).
- Selecting an even number from a standard die (3 out of 6).

Boundary: P = 1

Certain Event: Absolutely guaranteed to occur.

- Monday arriving immediately after Sunday.
- Picking a red sweet from a bag containing only red sweets!

Classroom Check: Ranking on the Scale

1. Where on the probability scale would you place the event: "An elephant will walk into your mathematics classroom today"?

   Assign a qualitative label and an approximate numeric value.

2. If the probability of your school winning tomorrow's match is 0.75, what is the probability that your school does NOT win?

   Notice that probabilities must sum to 1.

3. Can the probability of any real-world event ever be −0.2 or +1.5? Why or why not?

   Recall the absolute bounds of the scale: $0 \le P(E) \le 1$.

4. In a deck of 6 cards with 5 purple cards and 1 green card, how would you describe the likelihood of picking a purple card?

   Choose: Impossible, Less likely, Even chance, More likely, or Certain.

The Two Pillars of Objective Probability

1. Experimental (Empirical) Probability

Derived from Experience & Observation:

- Involves performing an experiment multiple times or gathering statistical data from past trials.
- Probability is calculated as the **relative frequency** of the event based on actual evidence.
- Does NOT assume that outcomes must be equal; works for irregular shapes (e.g., tossing a paper cup or thumb tack).

2. Theoretical (Classical) Probability

Derived from Logical Reasoning & Symmetry:

- Assumes an ideal, perfectly fair scenario where all possible outcomes are **equally likely**.
- Requires no physical experiment or past historical data.
- Calculated purely by counting favourable outcomes against the total number of equally likely outcomes.

Experience vs. Logic: Empirical Probability counts observed trials; Theoretical Probability deduces ideal symmetry.

Relative Frequency: Learning from Trials

MASTER FORMULA: EXPERIMENTAL PROBABILITY

Experimental Probability (Relative Frequency) = (Number of times the event occurred) / (Total number of trials conducted)

The 50-Roll Die Experiment (Example 2)

Suppose you roll a standard die 50 times and record the faces:

- Total number of trials = **50**.
- The number 4 appears exactly **8 times**.
- Experimental Probability of rolling a 4:
  P(4) = 8 / 50 = 0.16 (or 16%).
- The value 0.16 is the *relative frequency* of the event in this sample.

Irregular Objects: The Paper Cup (Fig. 7.5)

Theoretical symmetry does not exist for irregular shapes:

- When a paper cup is tossed into the air, it can land:
  - On its bottom
  - Upside down on its top
  - On its side
- These 3 outcomes are NOT equally likely!
- **Only experimental probability** (tossing it 100 times) can assign accurate probabilities to each landing position.

Classroom Check: Experimental Probability

1 A coin is tossed 20 times and lands on Heads 13 times. What is the experimental probability of getting Heads?

Compute the fraction: favourable occurrences / total tosses.

2 Why can we NOT use theoretical probability formulas to find the chance of a dropped drawing pin landing point-up?

Think about whether the outcomes are physically symmetrical.

3 In the 50-roll die experiment, why did the experimental probability (0.16) differ from the theoretical probability (1/6 ≈ 0.167)?

Consider the effect of a limited sample size.

4 If a paper cup lands on its side 74 times out of 100 tosses, what is the empirical probability of it landing on its side?

Express as both a decimal and a percentage.

Jñān-Chaupad: The Moral Origins of Dice Games

Ancient Indian Heritage

The global game of Snakes and Ladders:

- Originated in ancient and medieval India as **Jñān-Chaupad** (Game of Knowledge).
- Created as a pedagogical tool to impart philosophical and moral lessons.
- Illustrated on painted cloth scrolls and temple boards (preserved in the National Museum, New Delhi).

Ladders as Virtues

Symbolising spiritual ascent:

- Squares represented positive virtues: *Daya* (compassion), *Satya* (truth), *Dāna* (generosity), and *Jnāna* (wisdom).
- Landing on a virtue square propelled the player up the board toward spiritual liberation.

Snakes as Vices

Illustrating consequences of actions:

- Squares represented vices: *Krodha* (anger), *Lobha* (greed), and *Moha* (delusion).
- Landing on a vice caused the player to slide down, visually teaching that every choice carries consequences!

Theoretical Probability in Fair Systems

CLASSICAL DEFINITION OF PROBABILITY

For a fair experiment with equally likely outcomes:
P(Event) = (Number of favourable outcomes) / (Total number of possible outcomes) = n(E) / n(S)

Rolling a Standard Die (Example 3)

What is the probability of rolling a 4?

- Total possible outcomes: {1, 2, 3, 4, 5, 6} &implies; **n(S) = 6**.
- Favourable outcome: {4} &implies; **n(E) = 1**.
- **P(rolling a 4) = 1 / 6 ≈ 0.167 (or 16.7%)**.
- Notice: No die rolling was needed; logic and symmetry determine the answer!

A The Word 'PROBABILITY' (Example 4)

Pick a single letter at random from 'PROBABILITY':

- Total letters in the word = **11**.
- Count of the letter 'B' = **2** (P-R-O-**B**-A-**B**-I-L-I-T-Y).
- Favourable outcomes = 2.
- **P(picking letter B) = 2 / 11 ≈ 0.182 (or 18.2%)**.

Classroom Check: Theoretical Calculations

1 What is the theoretical probability of rolling an even number on a standard 6-sided die?

Count the even numbers: {2, 4, 6} out of 6.

2 A letter is chosen at random from the word 'MATHEMATICS'. What is the probability of picking the letter 'M'?

Count total letters and occurrences of M.

3 A standard pack of 52 cards has 4 Kings. What is the theoretical probability of drawing a King at random?

Simplify the fraction 4 / 52.

4 Why is the assumption of "equally likely outcomes" essential when applying the formula P(E) = n(E) / n(S)?

Consider what happens if one face of a die is weighted.

From Sample to Population: Real-World Estimation

Class Fruit Survey (Example 5)

Survey of 50 students' favorite fruits:

- Mango: 20 | Apple: 15 | Banana: 10 | Grapes: 5.
- **P(Mango)** = 20 / 50 = **0.4 (or 40%)**.
- Scaling to the whole school of **1500 students**:
  Expected Mangoes needed = 40% of 1500 = **600 mangoes**!
- Statistical probability allows organizations to forecast demands accurately.

Sample vs. Population Principles

Why sample quality matters:

- **Population:** The entire group under study (e.g., all 1500 students).
- **Sample:** The manageable subgroup actually surveyed (e.g., 50 students).
- **Sample Size:** Larger samples provide more reliable and confident estimates.
- **Representation & Bias:** The sample must represent all grades and genders fairly to prevent skewed results.

The Law of Large Numbers: Convergence of Trials

THE FUNDAMENTAL LAW OF PROBABILITY

As the number of trials in a fair random experiment increases toward infinity, the experimental probability (relative frequency) gets progressively closer to the theoretical probability!

Small Trials (N = 10)

- Fluctuations dominate.
- Tossing a coin 10 times might yield 8 Heads and 2 Tails ($P = 0.80$).
- Significant divergence from theoretical 0.50 is common and normal.

Moderate Trials (N = 100)

- Random spikes begin canceling out.
- Tossing 100 times might yield 54 Heads and 46 Tails ($P = 0.54$).
- The relative frequency steadily pulls toward the center.

Massive Trials (N = 10,000)

- Theoretical reality revealed.
- Tossing 10,000 times will yield extremely close to 5,000 Heads ($P \approx 0.5002$).
- *Probability does not predict what happens next; it predicts what happens in the long run!*

The Gambler’s Fallacy: Coins Have No Memory!

DEFINITION: GAMBLER'S FALLACY

The mistaken belief that if a random event occurs frequently in the short run, the opposite outcome is "due" or "more likely" to happen next to balance the score.

The Six-Heads Fallacy

Imagine tossing a fair coin that lands on Heads 6 times in a row:

- **The Flawed Thinking:** *"Tails hasn't appeared for so long; tails is bound to come next!"*
- **The Mathematical Truth:** The coin has zero memory! It does not keep a scorecard of past flips.
- On the 7th flip, P(Heads) = ½ and P(Tails) = ½, exactly the same as on the very first flip.

Snakes & Ladders Rolls (Example 6)

Rolling three consecutive 6s:

- A player thinks: *"I've rolled three 6s in a row; there is no way I can roll another 6 now!"*
- **Mathematical Reality:** Each dice roll is an **independent event**.
- The probability of rolling a 6 on the next turn is strictly **1 / 6 (16.7%)**, completely unchanged by history.

Classroom Check: Independence & Convergence

1 If a fair coin lands on Tails 8 times in a row, what is the probability that the 9th toss will be Heads?

Does past history influence the physics of the next independent flip?

2 What is the meaning of the mathematical statement: "A coin has no memory"?

Explain this in terms of independent events.

3 According to the Law of Large Numbers, what happens to the experimental probability of rolling a 3 as you roll a die 6000 times?

Compare the expected relative frequency to 1/6.

4 In a school survey of 40 students, 14 prefer Science Club. For a school of 800 students, how many would you estimate prefer Science Club?

Calculate (14 / 40) × 800.

Defining the Sample Space S and Sample Size n(S)

CORE RULES OF SAMPLE SPACE

The sample space S is the set containing all possible outcomes of an experiment. It must be exhaustive (no outcome missed) and mutually exclusive (no outcome listed twice).

Single-Step Experiments

- **Rain Experiment:**
  S = {Rain, No Rain} &implies; n(S) = 2.
- **Match Outcome:**
  S = {Win, Lose, Draw} &implies; n(S) = 3.
- **Single Die Roll:**
  S = {1, 2, 3, 4, 5, 6} &implies; n(S) = 6.

Tossing Two Coins Simultaneously

Order matters when tracking coins:

- Coin 1: H or T | Coin 2: H or T
- **S = {HH, HT, TH, TT}**
- Sample Size **n(S) = 4**.
- HT (Head on 1st, Tail on 2nd) is distinct from TH (Tail on 1st, Head on 2nd)!

Level of Detail (Page 167)

Think and Reflect:

- If weather requires nuanced measurement, the sample space expands:
- S = {No Rain, Drizzle, Light Rain, Heavy Rain} &implies; n(S) = 4.
- The sample space must match the precision required by the question!

Events: Favourable Subsets of the Sample Space

DEFINITION OF AN EVENT (E)

An event E is any specific collection of outcomes from the sample space. In set theory terms, an event is always a subset of the sample space ($E \subseteq S$).

Two-Coin Events

Sample Space S = {HH, HT, TH, TT}:

- **Event A ('At least one Head'):**
  E_1 = {HH, HT, TH} &implies; $n(E_1) = 3$.
  P(At least 1 Head) = 3 / 4.
- **Event B ('Both Tails'):**
  E_2 = {TT} &implies; $n(E_2) = 1$.
  P(Both Tails) = 1 / 4.

Die Roll Events

Sample Space S = {1, 2, 3, 4, 5, 6}:

- **Event C ('Number greater than 4'):**
  E_3 = {5, 6} &implies; $n(E_3) = 2$.
  P(Number > 4) = 2 / 6 = 1 / 3.
- **Event D ('Even Number'):**
  E_4 = {2, 4, 6} &implies; $n(E_4) = 3$.
  P(Even) = 3 / 6 = 1 / 2.

Fruit Basket Events

Sample S = {Apple, Banana, Orange}:

- **Event E ('Picking a yellow fruit'):**
  E_5 = {Banana} &implies; $n(E_5) = 1$.
- **Event F ('Picking a fruit with seeds'):**
  E_6 = {Apple, Orange} &implies; $n(E_6) = 2$.

Classroom Check: Sample Spaces & Subsets

1 What is the sample space S and sample size n(S) when choosing an integer strictly between −3 and +3?

List all integers: {-2, -1, 0, 1, 2}.

2 If a fair die is rolled, what subset represents the event "rolling a prime number"?

Identify the prime numbers in {1, 2, 3, 4, 5, 6}.

3 In a village fair offering Samosa, Pakora, and Bhaji with Chai or Lassi, how many total snack-drink combinations are possible?

Multiply choices: 3 snacks × 2 drinks.

4 Three coins are tossed. Why is {0, 1, 2} NOT a valid sample space for recording the number of heads?

What possible head count was omitted?

Tree Diagrams: Mapping Multi-Step Sequences

Anatomy of a Multi-Step Tree (Example 7)

Tossing a fair coin twice in succession (Fig. 7.6):

- **Stage 1 (First Toss):** Splits from start into 2 branches: H (prob ½) or T (prob ½).
- **Stage 2 (Second Toss):** From each branch, splits again into H (prob ½) or T (prob ½).
- **End Paths (Sample Space):**
  - Branch H → H: Outcome **HH** (Prob = ½ × ½ = ¼)
  - Branch H → T: Outcome **HT** (Prob = ½ × ½ = ¼)
  - Branch T → H: Outcome **TH** (Prob = ½ × ½ = ¼)
  - Branch T → T: Outcome **TT** (Prob = ½ × ½ = ¼)

Why Tree Diagrams Are Powerful

Key strategic advantages:

- **Zero Omissions:** Systematically ensures no possible path is accidentally forgotten.
- **Multiplicative Rule:** Multiplying branch probabilities along any path directly gives the probability of that joint outcome!
- **Handles Dependency:** Easily adapts to sampling *without replacement* by adjusting the second branch probabilities.

Classroom Check: Multi-Step Tree Diagrams

1 Using the two-coin tree diagram, what is the probability of getting exactly one Head and one Tail?

Identify which paths satisfy the event: {HT, TH}.

2 If you toss a coin three times, how many terminal branch outcomes will the complete tree diagram produce?

Calculate 2 × 2 × 2.

3 Basket A has 1 apple and 2 oranges. Basket B has 1 banana and 1 mango. How many branches connect the fruit selections?

Multiply options: 3 from Basket A × 2 from Basket B.

4 A box contains 4 red and 5 blue balls. If a ball is drawn and NOT replaced, what probability is written on the blue branch for the second draw?

Notice how the total denominator decreases to 8.

Geometric Probability: Continuous Area Ratios

PROBABILITY IN CONTINUOUS SPACE

When an object lands randomly within a continuous bounded region, the probability of it landing in a target zone is given by: P(Target) = (Area of Target Region) / (Total Area of Bounded Region).

The Liquid Dropper Problem (Fig. 7.8)

A drop of dye falls randomly onto a rectangular canvas:

- **Total Rectangular Region:** Dimensions 3 m × 2 m.
  Total Area = 3 × 2 = **6 m²**.
- **Target Circle:** Diameter = 1 m &implies; Radius r = 0.5 m = ½ m.
  Target Area = $\pi r^2 = \pi(\frac{1}{2})^2 = \mathbf{\pi / 4\ m^2}$.
- **P(Landing inside Circle)** = $(\pi / 4) / 6 = \mathbf{\pi / 24 \approx 0.1309}$ **(or 13.1%)**!

The 8-Sector Number Spinner (Fig. 7.7)

Spinning a fair circular dial with 8 equal sectors (numbered 1 to 8):

- Each numbered sector subtends an equal central angle of 360° / 8 = 45°.
- Total possible outcomes n(S) = 8.
- P(Arrow points at 8) = **1 / 8 = 0.125**.
- P(Odd Number: 1, 3, 5, 7) = **4 / 8 = ½ = 0.5**.
- P(Number > 2: 3, 4, 5, 6, 7, 8) = **6 / 8 = ¾ = 0.75**.

Classroom Check: Geometric & Spinner Probabilities

1 On the 8-numbered spinner (Fig. 7.7), what is the probability that the arrow points at a multiple of 3?

Identify the multiples of 3 among {1, 2, 3, 4, 5, 6, 7, 8}.

2 What is the probability that the arrow on the 8-numbered spinner lands on a number less than 9?

Is this an impossible, likely, or certain event?

3 In the rectangular dye dropper problem, what happens to the probability if the circle's diameter is doubled to 2 m?

Recall how area scales when diameter doubles.

4 When rolling a pair of standard 6-sided dice, give an example of an event that has a probability of exactly 0.

Think about the maximum possible sum of two dice.

Model Problem Solutions from Exercises

Tyre Durability (Ex Q7)

Given: 1000 tyre life records. Cases < 4000 km: 20; 4001–9000 km: 210; 9001–14000 km: 325; > 14000 km: 445.

- (i) P(< 4000 km) = 20 / 1000 = **0.02**.
- (ii) P(4000 to 14000 km) = (210 + 325) / 1000 = 535 / 1000 = **0.535**.
- (iii) P(> 14000 km) = 445 / 1000 = **0.445**.

The Word 'PEACE' (Ex Q8)

Given: Letters P, E, A, C, E on individual cards. Total cards = 5.

- (i) P(Letter is P, E, or C):
  Count = 1 (P) + 2 (E) + 1 (C) = 4 cards.
  P = 4 / 5 = 0.80.
- (ii) P(Letter is NOT an E):
  Count = 5 total – 2 (E) = 3 cards.
  P = 3 / 5 = 0.60.

Two Dice Prime Sum (Ex Q12(i))

Given: Two standard 6–sided dice are rolled simultaneously. Total outcomes = 6 × 6 = 36.

- Find P(Sum is prime and > 5):
  Possible prime sums > 5 are **7** and **11**.
- Sum 7: (1,6),(2,5),(3,4),(4,3),(5,2),(6,1) &implies; 6 pairs.
- Sum 11: (5,6),(6,5) &implies; 2 pairs.
- Total favourable = 6 + 2 = 8.
  P = 8 / 36 = 2 / 9 ≈ 0.222!

Master Probability Reference Checklist

⚖️ Principles & Scales

- ✅ **Definition:** Probability measures the likelihood of an event occurring under uncertainty.
- ✅ **Probability Scale:** Strictly bounded between 0 and 1: $0 \leq P(E) \leq 1$.
- ✅ **Boundary Conditions:** P = 0 denotes an Impossible event; P = 1 denotes a Certain event; P = 0.5 denotes an Even Chance.
- ✅ **Random Experiment:** Repeatable action where all outcomes are known, but single trials are unpredictable.
- ✅ **Independence:** Random events have no memory. The Gambler's Fallacy is mathematically false.

🧮 Calculation Models & Tools

- ✅ **Experimental Probability:** Relative Frequency = (Times occurred) / (Total trials conducted).
- ✅ **Theoretical Probability:** For equally likely outcomes: P(E) = n(E) / n(S).
- ✅ **Law of Large Numbers:** As trial counts increase, experimental frequency converges to theoretical probability.
- ✅ **Sample Space S:** Exhaustive set of all possible outcomes; Event E is a subset ($E \subseteq S$).
- ✅ **Tree Diagrams:** Map multi-step sequential trials and multiply branch probabilities.
- ✅ **Geometric Probability:** Continuous target area divided by total boundary area.

END OF CHAPTER 7

THANK YOU

"Probability is the very guide of life — it transforms human uncertainty into mathematical clarity, teaching us that while the next moment is a mystery, the pattern of the universe is rational."

✦ Akash Srivastva (www.akashmaths.online)